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FACULTÉ DES SCIENCES
DÉPARTEMENT D'INFORMATIQUE

Digital Twin for railway network simulation

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Master's thesis in Computer Science
Academic year 2025–2026

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Chapter 1

Introduction

Railway networks play an essential role in modern transportation systems. They support the daily mobility of passengers, contribute to freight transportation, and constitute a strategic infrastructure for sustainable mobility. In dense railway networks such as the Belgian one, the ability to operate trains efficiently, safely, and reliably is particularly important. A large number of trains, stations, tracks, and operational constraints interact continuously, making railway traffic management a complex task.

This complexity is not only due to the size of the infrastructure, but also to the nature of railway operations. Trains must follow predefined routes and timetables, share limited track capacity, respect safety constraints, and interact with infrastructure elements such as switches, platforms, and stations. A small disruption in one part of the network can propagate to other trains and stations, affecting punctuality and service quality. As a result, railway operators and researchers increasingly rely on digital tools to better understand, simulate, and analyse railway systems.

In this context, the concept of a digital twin has gained increasing attention. Informally, a digital twin can be understood as a virtual counterpart of a real physical system. It is not only a static model or a simple visualization of the system, but rather a digital representation that can be connected to data, updated over time, and used to observe, simulate, and analyse the behaviour of the physical system. Applied to railways, a digital twin may represent parts of the railway infrastructure, train movements, operational constraints, and traffic conditions. Such a representation can help explore how the network behaves, test different scenarios, and support decision-making.

Simulation is a central component of this vision. Before a railway digital twin can be used for prediction, optimization, or real-time decision support, it must first be possible to construct a coherent simulation model from available infrastructure and operational data. Simulation makes it possible to reproduce train movements in a controlled environment, evaluate whether generated routes are feasible, and compare simulated behaviour with real railway data.

This thesis focuses on the construction of a simulation-oriented railway digital twin using SUMO. SUMO is a traffic simulation platform originally designed for road traffic, but its flexible network and route representation make it possible to adapt it to railway use cases. The work investigates how railway infrastructure and operational data can be transformed into simulation-ready artefacts, and how different levels of railway network representation can be used to support train simulation.

1.1 Problem Statement

Despite the availability of open data sources and simulation platforms, constructing a realistic railway simulation remains a difficult task. Railway data is heterogeneous by nature: infrastructure data describes physical components such as stations, tracks, switches, and platforms, while operational data describes train movements, schedules, and punctuality records. These datasets do not always share the same structure, granularity, or level of detail. Consequently, integrating them into a single simulation framework requires several processing, modeling, and approximation steps.

A first difficulty concerns the representation of the railway infrastructure. At a high level, a railway network can be represented as a graph where stations are connected by railway links. This abstraction is useful because it simplifies the network and makes it easier to reconstruct train movements between stations. However, it does not represent detailed infrastructure elements such as platforms, switches, parallel tracks, or station-level routing constraints. These details are important when the objective is to simulate train circulation more realistically.

A second difficulty appears when moving toward a more detailed representation of the network. At the micro level, tracks, platforms, and switches must be represented explicitly. This provides a more realistic description of the infrastructure, but it also introduces new challenges. The network must be topologically consistent, platforms must be associated with the correct tracks, and valid routes must exist between consecutive stations. Even small errors in the reconstructed infrastructure can prevent trains from being routed correctly.

A third difficulty concerns the integration of real operational data. Punctuality data provides information about train stops, arrival times, departure times, and delays, but it does not directly provide the exact physical route followed by each train inside the infrastructure. Therefore, a route must be reconstructed from the available station sequences and mapped onto the generated simulation network. When the network is simplified, this process is easier but less precise. When the network is detailed, the process becomes more realistic but also more fragile.

These challenges lead to the central research question of this thesis:

How can a railway digital twin simulation be constructed using SUMO and heterogeneous railway datasets, and to what extent can real operational data be integrated into feasible macro-level and micro-level railway simulations?

This question is addressed by studying two complementary modeling approaches. The first one is a macro-level model, where stations are represented as nodes and station-to-station connections as edges. The second one is a micro-level model, where detailed infrastructure elements such as tracks, switches, and platforms are explicitly represented. Together, these two approaches make it possible to evaluate the trade-off between simplicity, scalability, infrastructure realism, and simulation feasibility.

1.2 Objectives

The main objective of this thesis is to design and implement a data-driven pipeline for constructing a simulation-oriented digital twin of a railway network. The focus is not

to build a fully operational real-time digital twin, but rather to study the foundations required to generate, simulate, and evaluate railway traffic scenarios from real-world data.

More specifically, this work aims to construct railway simulation models at two different levels of abstraction. The macro-level model is designed to represent the railway network as a station-to-station graph. Its purpose is to provide a simplified but functional simulation environment in which real train trajectories can be reconstructed from punctuality data. This model serves as a first validation of the complete pipeline, from data preprocessing to simulation execution.

The second objective is to develop a micro-level model that represents the infrastructure in greater detail. This includes the processing of track data, the detection of switches, the identification of platforms, the segmentation of track sections, and the construction of routing structures. The objective of this model is to move closer to the physical structure of the railway network and to evaluate whether more detailed simulations can be produced from the available data.

Another objective is to compare different trip generation strategies. In the macro-level model, the focus is mainly on the use of real operational data to reconstruct train movements. In the micro-level model, two approaches are considered: synthetic trip generation and real-data-based trip generation. Synthetic trips are used to test the feasibility and connectivity of the generated infrastructure, while real-data trips aim to reproduce observed railway operations more accurately.

Finally, this thesis aims to identify the main limitations encountered when building a railway digital twin simulation from heterogeneous data. These limitations concern data availability, infrastructure completeness, modeling assumptions, routing feasibility, and constraints imposed by the simulation platform. By analysing these issues, the work provides methodological insights for future developments of railway digital twins.

The expected outcome of this thesis is therefore not a perfect reproduction of the Belgian railway network, but a structured and evaluated prototype demonstrating how railway data can be transformed into SUMO-compatible simulation models. The contribution lies in the development of the modeling pipeline, the comparison between macro-level and micro-level representations, and the identification of the technical challenges that must be addressed to progress toward more complete railway digital twins.

1.3 Thesis Structure

The remainder of this thesis is organized as follows.

Chapter 2 presents the state of the art. It introduces the concept of digital twins, first from a general perspective and then in the context of railway systems. It also reviews the main components of railway infrastructure and operations, discusses common approaches for railway network representation, and presents existing railway simulation tools. Particular attention is given to SUMO and to its suitability for railway simulation.

Chapter 3 describes the proposed methodology. It positions the work with respect to the research gaps identified in the literature and presents the overall framework followed throughout the project. This chapter also justifies the choice of SUMO as the simulation platform and describes the infrastructure and operational datasets used in the implementation.

Chapter 4 details the implementation of the proposed approach. It is divided into two main parts. The first part presents the macro-level model, including data processing, network generation, trip generation, simulation, and its limitations. The second part

presents the micro-level model, including infrastructure modelling, switch detection, platform detection, track segmentation, dual graph construction, trip generation, simulation, and its limitations.

Chapter 5 presents the results obtained from the implemented models. It analyses dataset coverage, infrastructure modelling quality, simulation results, and computational performance. The objective of this chapter is to evaluate how well the generated models represent the available railway data and how effectively they can be used for simulation.

Chapter 6 discusses the main limitations of the proposed methodology. While some limitations are briefly introduced in the implementation chapter, this chapter provides a more detailed analysis of issues related to data quality, modelling assumptions, SUMO constraints, and the generalization of the approach to other railway networks.

Chapter 7 concludes the thesis by summarizing the main contributions and findings. It also discusses the practical implications of the work and presents future research directions, including counterfactual simulation, track work planning, real-time forecasting, incident propagation, and the generalization of the proposed framework.

Chapter 2

State of the Art

2.1 Digital Twins

Digital twins have become one of the central concepts associated with the digitalization of complex industrial systems ([3]). The term is used in many domains, including manufacturing, aerospace, healthcare, smart cities, energy systems, and transportation ([3, 9, 8]). Although the exact definition may vary depending on the application domain, the general idea remains the same: a digital twin is a digital representation of a physical system that is connected to data and used to analyse, simulate, monitor, or support decisions about the real system ([5, 3]).

In the context of this thesis, the concept of a digital twin is particularly relevant because railway networks are complex, dynamic, and data-rich systems. They involve physical infrastructure, moving vehicles, operational rules, timetables, delays, maintenance constraints, and safety requirements. A digital twin can therefore provide a structured framework to combine infrastructure data, operational data, simulation models, and analytical tools into a coherent representation of the railway system.

2.1.1 Definition, Capabilities and Limitations

The concept of digital twin was initially introduced in the context of product lifecycle management and later expanded to other engineering and industrial domains. A commonly accepted definition describes a digital twin as a virtual representation of a physical object, process, or system that is connected to its physical counterpart through data exchange and that can be used throughout the lifecycle of the system. This definition emphasizes that a digital twin is not merely a static model or a visualization tool. Its distinguishing characteristic is the relationship between the physical entity and its digital counterpart.

A digital twin can therefore be understood as the combination of three fundamental components: the physical system, the virtual system, and the connection between them. The physical system corresponds to the real-world object or process being represented. The virtual system is the digital model that describes the structure, behaviour, and state of the physical system. The connection corresponds to the data flow that links both entities. This connection may be unidirectional, when the physical system only updates the virtual model, or bidirectional, when the digital model can also influence decisions or actions applied to the physical system.

Figure 2.1: General architecture of a digital twin.

A more formal way to describe a digital twin is to represent it as a system composed of several interacting elements ([11, 3]):

$$DT = (P, V, D, C, S) \quad (2.1)$$

where P denotes the physical system, V the virtual representation, D the data associated with the system, C the connection mechanisms between the physical and virtual entities, and S the services built on top of the twin, such as simulation, prediction, visualization, diagnosis, or optimization. This representation highlights that the digital twin is not only a model, but an architecture in which data, models, and services are combined to support decision-making.

This distinction is important because the term digital twin is sometimes used broadly to describe any digital model of a real system. However, several authors distinguish between a digital model, a digital shadow, and a digital twin ([7]). A digital model is a digital representation of a system without automatic data exchange with the physical system. A digital shadow receives data from the physical system, but the information flow is mainly one-way. A digital twin represents a more advanced form, where the digital representation is continuously or regularly updated from the physical system and can support feedback, analysis, or operational decisions ([7]). In this thesis, the proposed work is closer to a simulation-oriented digital twin foundation: it focuses on constructing a coherent virtual railway representation from real infrastructure and operational data, which is a necessary step before reaching a fully real-time and bidirectional digital twin.

Digital twins are usually associated with several capabilities. The first is monitoring, which consists of observing the state of a system using data collected from sensors, logs, operational records, or information systems. The second is simulation, where the digital representation is used to reproduce the behaviour of the physical system under controlled conditions ([3, 10]). The third is prediction, where historical and real-time data are used to estimate future states, failures, delays, or performance indicators ([3, 9, 8]). The fourth is optimization, where alternative scenarios are evaluated in order to support better decisions. Finally, more advanced digital twins may support control or actuation, where decisions derived from the digital model influence the operation of the physical system.

These capabilities make digital twins particularly attractive for complex systems, but they also introduce significant limitations. A digital twin is only as reliable as the data, models, assumptions, and synchronization mechanisms on which it is based. If the data is incomplete, outdated, noisy, or inconsistent, the digital twin may provide misleading results. Similarly, if the model is too simplified, it may fail to reproduce important physical or operational constraints. On the other hand, if the model is too detailed, it may become computationally expensive and difficult to maintain.

Another limitation concerns validation. Since a digital twin is expected to represent a real system, it must be evaluated against real observations. This is challenging because real systems are often affected by hidden variables, incomplete measurements, operational exceptions, and human decisions that are not always captured in the available data. For this reason, the development of a digital twin is usually incremental. A first version may focus on static representation and simulation, while later versions progressively integrate real-time data, predictive models, and optimization services.

2.1.2 Recent Developments and Applications

The development of digital twins has accelerated with the progress of several enabling technologies. The IoTs allows physical systems to generate continuous streams of data ([3]). Cloud computing and distributed architectures provide the infrastructure required to store and process large volumes of information. Artificial intelligence and machine learning make it possible to extract patterns, predict future states, and detect anomalies. Simulation tools provide an environment in which alternative scenarios can be tested without disrupting the real system. Finally, visualization technologies, including dashboards, 3D interfaces, and geographic information systems, help users interpret the state and behaviour of the DT.

Digital twins have first been widely adopted in manufacturing, where they support product design, production monitoring, maintenance planning, and process optimization. In smart manufacturing, a digital twin can represent a machine, a production line, or an entire factory. It can be used to monitor equipment status, simulate production scenarios, predict failures, and improve resource allocation ([10]). This domain is particularly important in the literature because it provides many of the conceptual foundations later reused in other fields.

In aerospace and mechanical engineering, digital twins are used to monitor the condition of critical assets and support predictive maintenance. Aircraft, engines, turbines, and other complex mechanical systems generate large quantities of operational data during their lifecycle. A digital twin can integrate this data with physics-based or data-driven models in order to estimate wear, detect abnormal behaviour, and plan maintenance interventions before failures occur. Similar principles are found in energy systems, where digital twins are used for turbines, power plants, and renewable energy infrastructure ([8]).

Digital twins are also increasingly studied in healthcare and smart cities. In healthcare, the objective may be to represent medical devices, hospital workflows, or even patient-specific physiological models. In smart cities, digital twins can integrate mobility data, infrastructure data, environmental data, and urban services to support planning and decision-making. These examples show that digital twins are not limited to individual physical objects. They can also represent complex systems composed of many interacting components.

Across these application domains, digital twins generally serve four main purposes. First, they improve visibility by providing a structured representation of a system that may otherwise be difficult to observe directly. Second, they support experimentation by allowing users to test what-if scenarios in a virtual environment. Third, they improve anticipation by enabling predictive analysis based on past and current data. Fourth, they support decision-making by comparing alternative actions before they are applied to the real system.

However, the maturity of digital twin implementations varies significantly. In some cases, the term digital twin refers to a highly connected and dynamic system with real-time data exchange. In other cases, it refers to a simulation model or a dashboard that is only periodically updated. This variation shows that the digital twin concept should not be treated as a single technology, but rather as a framework that can be implemented at different levels of maturity. For this thesis, the focus is placed on the simulation and data integration layers, which are essential prerequisites for more advanced digital twin capabilities such as real-time forecasting, optimization, and decision support.

2.1.3 Applications in Railway networks

Railway systems are particularly suitable candidates for digital twin approaches. They combine physical infrastructure, operational processes, moving assets, strict safety constraints, and large volumes of data. A railway network includes tracks, switches, signals, stations, platforms, rolling stock, timetables, passenger flows, maintenance operations, and traffic management rules. These elements interact continuously, and a local disturbance can propagate through the network. As a result, railway operators require tools that can represent the current state of the network, simulate traffic behaviour, evaluate disruptions, and support planning decisions.

Digital twins for railway networks can be applied at different levels of granularity. At the asset level, a digital twin can represent individual components such as tracks, switches, bridges, tunnels, trains, or signalling equipment. This type of twin is particularly useful for condition monitoring and predictive maintenance. Sensor data, inspection records, and maintenance logs can be used to estimate the health of the asset and identify when intervention is required.

At the infrastructure level, a digital twin can represent the topology of the railway network. This includes the spatial structure of tracks, stations, platforms, junctions, and operational segments. Such a representation is necessary to analyse accessibility, capacity, routing possibilities, and infrastructure conflicts. In this thesis, this level is central because the objective is to transform real railway infrastructure data into simulation-ready network models.

At the operational level, a railway digital twin can represent train movements, schedules, delays, and traffic interactions. This makes it possible to analyse how trains circulate through the network, how delays propagate, and how infrastructure constraints affect service quality. A simulation-based digital twin can also be used to evaluate what-if scenarios, such as modified timetables, temporary track closures, traffic disruptions, or infrastructure works.

Figure 2.2: Example of a railway digital twin architecture.

Several applications can be identified in the railway domain. The first is predictive maintenance. Railway infrastructure is expensive to inspect and maintain, and failures can have significant consequences on safety and service reliability. A digital twin can combine sensor measurements, historical degradation data, and maintenance records to anticipate failures and optimize maintenance planning. The second application is traffic management. By reproducing train movements in a virtual environment, a digital twin can help evaluate congestion, detect bottlenecks, and test operational strategies. The third application is delay analysis and propagation modelling. Since railway operations are strongly interconnected, a delay affecting one train can influence many others. A digital twin can help identify where delays originate and how they spread across the network.

A fourth application is scenario testing. Railway operators often need to evaluate changes before implementing them in the real network. Examples include timetable modifications, station layout changes, temporary closures, track works, or emergency response strategies. A simulation-based digital twin makes it possible to test such scenarios in a controlled environment and compare their expected impact. This capability is especially relevant for this thesis, since the proposed approach focuses on generating railway simulations from real infrastructure and operational data.

Railway digital twins also raise specific challenges. Railway networks are safety-critical systems, which means that simplified models must be used with care. Accurate representation of tracks, switches, signals, platforms, and routing constraints is necessary when the objective is to reproduce detailed operational behaviour. Data availability is another major challenge. Infrastructure data may be incomplete, private, heterogeneous, or difficult to align with operational data. Finally, railway systems operate at different spatial and temporal scales. A digital twin may need to represent long-distance network connectivity, station-level infrastructure, and real-time traffic dynamics at the same time.

For these reasons, railway digital twins often require a layered approach. A macro-level representation can be used to model station-to-station connectivity and global train movements. A micro-level representation can then be used to represent detailed infrastructure such as tracks, switches, and platforms. This distinction is directly reflected in the methodology of this thesis, where both macro-level and micro-level railway models are developed. The macro-level model provides a simplified and scalable representation of the network, while the micro-level model aims to capture more detailed infrastructure constraints. Together, these two levels contribute to the construction of a simulation-oriented foundation for a railway digital twin.

In summary, digital twins provide a relevant conceptual and technical framework for railway simulation. They offer a way to combine infrastructure data, operational data, simulation models, and analytical tools into a unified system. However, building a complete railway digital twin requires several intermediate steps: reconstructing the infrastructure, generating valid train routes, integrating real operational data, validating simulation outputs, and identifying the limitations of the model. The work presented in this thesis focuses on these foundational steps, with the objective of preparing the ground for future digital twin applications such as real-time forecasting, disruption analysis, and decision support.

2.2 Railway Systems

2.2.1 Railway infrastructure

Tracks

Switches and Junctions

Stations and Platforms

Signaling Systems

2.2.2 Railway Operations

Train Movements

Timetables

Delay Propagation

2.2.3 Railway Network Representation

Graph-Based Modeling

Physics-Based Modeling

Logic-Based Modeling

2.2.4 Railway Data Sources

Infrastructure Data

Operational Data

2.3 Railway Simulators

2.3.1 Overview of Existing Simulators

2.3.2 Comparative Analysis

2.3.3 Limitations and Gaps in Current Simulators

2.3.4 SUMO: General Features

2.3.5 SUMO Backend Architecture

2.3.6 SUMO For Railway Simulation

2.4 Simulation Calibration

2.5 Research Gaps

Chapter 3

Proposed Methodology

This chapter presents the methodology proposed in this thesis to construct a simulation-oriented digital twin of a railway network. The objective is to define and implement the foundational components required to transform heterogeneous railway datasets into simulation-ready models. These components include infrastructure reconstruction, operational data integration, trip generation, simulation execution, and result validation.

The proposed methodology is structured around two complementary levels of representation. The first one is a macro-level model, in which the railway network is represented as a station-to-station graph. This model provides a simplified but scalable representation of the railway system and is mainly used to reconstruct and simulate train movements from real operational data. The second one is a micro-level model, in which the railway infrastructure is represented in greater detail through tracks, switches, platforms, and operational segments. This model aims to move closer to the physical structure of the railway network and to evaluate whether more detailed train simulations can be generated from the available data.

This dual-level approach makes it possible to study the trade-off between simplicity and realism. The macro-level model is easier to construct, more robust for route reconstruction, and better suited for validating the end-to-end workflow. However, it cannot represent local infrastructure constraints inside stations. The micro-level model provides a richer infrastructure description, but it also introduces additional challenges related to data completeness, platform assignment, switch detection, route feasibility, and simulation complexity. By developing both approaches, the methodology evaluates how different levels of abstraction influence the construction of a railway digital twin.

3.1 Positioning of This Work

3.2 Framework

The proposed framework follows a data-driven pipeline that transforms raw railway datasets into SUMO-compatible simulation artefacts. The general architecture can be divided into five main stages: data collection, data preprocessing, infrastructure modeling, trip generation, and simulation-based evaluation. These stages are common to both the macro-level and micro-level approaches, but their implementation differs according to the level of detail required by each model.

The first stage consists of collecting the available railway data. Two main categories

of data are considered: infrastructure data and operational data. Infrastructure data describes the physical and topological structure of the railway network, including stations, station-to-station connections, tracks, platforms, and switches. Operational data describes train movements, timetables, arrival and departure times, and punctuality information. These datasets provide the basis for constructing the virtual representation of the railway system.

The second stage corresponds to data preprocessing. Since railway data is heterogeneous, it must be cleaned, standardized, and transformed before it can be used for simulation. This includes harmonizing station identifiers, processing geographical coordinates, filtering the data according to the selected simulation scope, and converting temporal information into simulation-compatible time values. This step is essential because inconsistencies between infrastructure and operational datasets can prevent route reconstruction or lead to invalid simulations.

The third stage is infrastructure modeling. In the macro-level model, the infrastructure is represented as a graph in which stations correspond to nodes and direct railway connections correspond to edges. This representation is suitable for station-to-station simulation and for reconstructing train trajectories from punctuality data. In the micro-level model, the infrastructure is reconstructed at a finer level of detail. Individual tracks are represented explicitly, switches are detected from network topology, platforms are matched to nearby track segments, and the network is segmented when necessary to improve the operational granularity of the model.

The fourth stage is trip generation. In the macro-level model, train trips are reconstructed from real punctuality records. Each train journey is obtained by grouping station records by train identifier and operating day, then sorting them according to their planned times. The resulting station sequences are converted into SUMO routes. In the micro-level model, two trip generation strategies are considered. The first one generates random trips constrained by the detected platforms and network topology. The second one attempts to reconstruct real train journeys from operational data and map them onto the detailed infrastructure.

The final stage is simulation and evaluation. Once the network and routes have been generated, the scenario is executed in SUMO. The simulation produces output files describing train movements, stops, travel times, and summary statistics. These outputs are then processed and compared with the available reference data. The evaluation focuses on dataset coverage, infrastructure modeling quality, and the feasibility of the generated simulations. The framework therefore provides both a technical implementation pipeline and an evaluation methodology for assessing the quality of the generated railway digital twin.

3.3 Why SUMO?

3.4 Data Available

The proposed methodology relies on two main types of datasets: infrastructure data and operational data. Infrastructure data is used to construct the virtual representation of the railway network, while operational data is used to generate train movements and evaluate whether real railway activity can be reproduced in simulation. The use of both categories is essential because a railway digital twin must represent not only the physical structure

of the network, but also the way trains actually operate on it.

The data used in this thesis comes from Infrabel Open Data Portal And DTSC who has access to private data directly from Infrabel. These datasets are not directly simulation-ready. They must be cleaned, aligned, transformed, and sometimes approximated before they can be converted into SUMO artefacts. The available data also differs depending on the level of representation. The macro-level model mainly uses stations and station-to-station connections, while the micro-level model requires more detailed information about tracks, platforms, and switches.

3.4.1 Infrastructure Data

Infrastructure data corresponds to the static elements of the railway network. It includes information about stations, connections between stations, railway track geometries, platform locations, and switch-like topological structures. In this thesis, infrastructure data is used to generate the SUMO network on which train movements are simulated.

The use of infrastructure data differs between the macro-level and micro-level approaches. The macro-level model only requires stations and direct connections between them. It therefore focuses on the general connectivity of the railway network. The micro-level model requires more detailed infrastructure elements and aims to represent the physical topology of the railway network more closely.

Macro-Level Data

The macro-level infrastructure data represents the railway system as a graph. In this graph, stations are modeled as nodes and direct railway connections between stations are modeled as edges. This abstraction does not include detailed elements such as platforms, switches, signals, or internal station layouts. Instead, it focuses on preserving the overall topology of the railway network.

This level of data is particularly useful for reconstructing train movements from station sequences. Since the operational data contains train passages through stations, the macro-level representation makes it possible to map these station sequences directly onto a simulation network. The objective is therefore not to reproduce detailed infrastructure conflicts, but to build a robust and scalable representation of railway circulation between stations.

Dataset		Example tributes	at- Reference value	Role
Stations		Station identifier, station name, lati- tude, longitude	559 stations	Used to create SUMO nodes rep- resenting railway stations
Stations Connections	Connec- tions	Origin station, destination station, distance, geometry	1385 tracks	Used to create SUMO edges be- tween neighbouring stations
Selected Area	Simulated Area	List of selected stations or station chain	User-defined	Used to restrict the simulation to a co- herent part of the network

Table 3.1: Macro-Level Infrastructure Data Overview

The macro-level data therefore provides the minimal infrastructure required to simulate trains from one station to another. Its main advantage is that it reduces the complexity of the railway system while keeping the information needed to reconstruct real train trajectories.

Micro-Level Data

The micro-level infrastructure data provides a more detailed representation of the railway network. It considers physical railway elements such as track segments, platforms, and switches. This level of representation is required when the objective is to simulate train circulation closer to the real infrastructure.

Track data is used to generate the base railway graph. Each track segment contains geometric and topological information that makes it possible to construct the SUMO network. Platform data is used to identify where trains can stop inside stations. Since platforms must be associated with SUMO lanes, a matching step is required to link each platform to a nearby track segment. Switches are not used only as isolated objects, but also as topological indicators that help describe how track segments are connected.

An example of the type of information used at the micro level is shown in Table 3.2.

Dataset	Example tributes	at- Reference value	Role
Tracks segments	Track identifier, geometry, start point, end point, length, geometry	25068 tracks	Used to construct detailed SUMO railway edges
Platforms	Station name, platform number, associated area	1560 platforms	Used to create train stops in the SUMO additional file
Switches	Node identifier, coordinates, connected track segments	10379 switches	Used to represent branching points and support routing through the network

Table 3.2: Micro-Level Infrastructure Data Overview

3.4.2 Operational Data

Operational data describes train movements across the railway network. In this thesis, this data is mainly used to reconstruct train journeys and to generate simulation routes. Each record corresponds to the passage of a train at a given station and contains temporal information related to the planned and observed operation of the service.

The operational dataset is essential because it provides a connection between the static infrastructure and real railway activity. Without operational data, the generated network would only represent the physical railway system, but it would not contain realistic train movements. By using train identifiers, station identifiers, and planned times, it becomes possible to reconstruct ordered sequences of stops for each train.

An example of the type of information contained in the operational data is shown in Table 3.3.

Dataset	Main tribute groups	at-	Example tributes	at-	Reference value	Role
Punctuality dataset	Train identification, station pas- sage, timetable information, observed op- eration, delay information	iden-	Train identi- fier, operating date, station identifier, planned arrival time, planned departure time, observed ar- rival/departure time, arrival delay, depar- ture delay		Depends on selected time window	Used to re- construct real train journeys, generate sim- ulation routes, and compare simulated operations with observed railway move- ments

Table 3.3: Operational Data Overview

At the macro level, the operational data can be mapped relatively directly onto the station-to-station graph. For each train and each operating day, the records are grouped and sorted chronologically to obtain a sequence of visited stations. This sequence is then transformed into a SUMO route.

At the micro level, the same operational data is more difficult to use because each station stop must be associated with a specific platform or track segment. In addition, the route between two consecutive stations must be computed through the detailed infrastructure graph. The operational data therefore remains the same in principle, but its integration becomes more constrained when the level of infrastructure detail increases.

Overall, the available data provides the foundation for constructing the proposed railway digital twin. Infrastructure data defines the network to be simulated, while operational data defines the train movements to be reproduced. The distinction between macro-level and micro-level data is central to the methodology, because it determines the type of simulation that can be generated and the complexity of the preprocessing steps required before execution in SUMO.

Chapter 4

Implementation

4.1 Macro-Level Modeling

The macro-level model constitutes the first operational implementation of the railway digital twin developed in this work. Its primary objective is to generate a simulation-ready representation of the Belgian railway network while maintaining a level of abstraction that remains computationally manageable. Unlike the micro-level model presented later in this thesis, which aims to reproduce the detailed infrastructure of railway stations and track layouts, the macro-level model focuses on the connectivity between stations and the movement of trains across the network.

At this level of abstraction, railway stations are represented as nodes and railway connections between stations are represented as edges. Detailed infrastructure components such as switches, signaling systems, individual tracks, and platform assignments are intentionally omitted. This simplification significantly reduces the complexity of the generated network while preserving its overall topology and operational structure.

The implementation of the macro-level model serves several purposes. First, it validates the complete workflow required to construct a railway digital twin, from raw infrastructure and operational data to simulation outputs. Second, it provides an initial environment for evaluating trip generation methods based on real railway operations. Finally, it establishes a baseline model against which the more detailed micro-level implementation can later be compared.

The implementation was developed in Python and relies on Apache Spark for large-scale data processing and SUMO for railway simulation. The complete workflow consists of four main stages: data processing, railway network generation, trip generation, and simulation execution.

4.1.1 Data Processing Pipeline

The generation of the macro-level railway model begins with the construction of a data processing pipeline responsible for transforming heterogeneous railway datasets into a format compatible with SUMO. Railway data originates from a unique source but exhibits significant differences in structure, granularity, and purpose. Consequently, a preprocessing phase is required before any simulation artefacts can be generated.

Three categories of data are used within the macro-level model:

- **Railway station data:** contains information describing the geographical location and identification of Belgian railway stations.

- **Railway connectivity data:** provides information regarding the direct railway links connecting stations.
- **Railway punctuality data:** operational records describing the observed movements of trains across the railway network.

The preprocessing workflow follows a classical Extract-Transform-Load (ETL) approach. During the extraction phase, raw datasets are loaded into Apache Spark dataframes. Spark was selected because of its ability to efficiently process large datasets while providing a unified interface for data transformation operations. Although the macro-level simulation itself does not require distributed execution, the volume of punctuality records makes Spark particularly suitable for preprocessing tasks.

Once loaded, the datasets undergo a transformation phase during which inconsistencies are corrected and the data structure is standardized. Particular attention is paid to geographical information. Infrastructure datasets and operational datasets do not always use identical coordinate representations, resulting in small discrepancies between station locations and railway line geometries. Coordinate normalization procedures are therefore applied to ensure consistency between datasets.

A filtering phase is subsequently performed to restrict the simulation to the geographical area and time period selected by the user. The simulation generator allows a subset of stations to be specified as input. In addition to these stations, neighboring stations required to preserve network connectivity are automatically included in the simulation. This approach reduces computational complexity while ensuring that realistic routes can still be generated.

The output of the preprocessing pipeline consists of a collection of cleaned datasets containing standardized station information, station-to-station railway connections, and train circulation records. These datasets constitute the foundation upon which the simulation network is generated.

4.1.2 Railway Network Generation

Once the preprocessing phase has been completed, the railway network can be generated. The objective of this stage is to transform the cleaned infrastructure datasets into a graph representation that can subsequently be converted into a SUMO railway network.

The process begins with the loading of station information. Each station is uniquely identified by its official identifier and associated with a geographical position. Internally, station information is stored in dictionary-based structures that provide efficient access during subsequent processing stages. For each station, the implementation stores its identifier, name, and geographical coordinates.

After station loading, railway links between stations are reconstructed from the station-to-station dataset. Each railway connection contains the identifier of the departure station, the identifier of the arrival station, the physical distance separating the two stations, and the geographical shape describing the railway corridor.

Unlike a purely abstract graph representation, the implementation preserves the geometry of railway lines whenever this information is available. Railway corridors are therefore represented as polylines rather than simple straight segments. This decision improves the visual realism of the generated SUMO network while preserving the simplicity of the macro-level model.

The resulting railway network can be interpreted as a directed graph $G = (V, E)$, where the set of vertices V corresponds to railway stations and the set of edges E corresponds to direct railway connections between stations.

Once the graph has been constructed, a series of XML files compatible with SUMO are generated automatically. These files include the node definitions, edge definitions, stop definitions, train definitions, and simulation configuration files required to execute the simulation.

The generated node file contains all stations included in the simulation. Each station is represented as a SUMO node associated with a unique identifier and geographical coordinates. Similarly, the edge file contains the railway links connecting stations and specifies the geometry and characteristics of each connection.

The final step consists of converting these files into a SUMO network using the `netconvert` utility. This tool automatically transforms the graph representation into a simulation-ready railway network. The resulting network preserves the geographical distribution of stations and railway corridors while remaining significantly simpler than the actual Belgian railway infrastructure.

This abstraction offers several advantages. The resulting network is easy to generate, computationally efficient, and sufficiently realistic for validating trip generation and operational workflows. However, it also introduces limitations. Since stations are represented by a single node, infrastructure-level phenomena such as route conflicts, switch occupation, signal interactions, and platform assignment cannot be reproduced accurately.

4.1.3 Trip Generation

Once the railway network has been generated, train movements must be created in order to execute the simulation. Rather than relying exclusively on synthetic traffic generation, the proposed methodology aims to reconstruct train movements from real operational data.

The punctuality dataset contains records describing successive train passages through railway stations. Each record includes a train identifier, a station identifier, and temporal information such as planned departure times and observed departure times. Although these records provide valuable operational information, they do not directly describe complete train journeys.

A reconstruction process is therefore required to transform isolated station observations into coherent train routes. For each train and operating day, records are first sorted chronologically according to their planned departure time. The next station visited by the train is then identified using window-based operations implemented in Apache Spark.

This procedure allows consecutive station pairs to be extracted and transformed into station-to-station movements. The resulting dataset can be interpreted as a collection of train trajectories describing the progression of trains through the railway network.

For each reconstructed movement, a simulation departure time must be computed. Since SUMO internally represents time as the number of seconds elapsed since the beginning of the simulation, departure times observed in the punctuality dataset are converted into simulation timestamps. This conversion is performed by calculating the difference between the planned departure time and the simulation start date selected by the user.

The reconstructed train movements are subsequently aggregated into complete train journeys. Each journey contains the train identifier, the sequence of visited stations, and the associated departure times. These journeys are then translated into SUMO route

definitions.

The use of real operational data offers several advantages over purely random traffic generation. First, it preserves realistic traffic density patterns. Second, it reproduces actual station usage frequencies and train departure distributions. Finally, it provides a more representative basis for evaluating the behaviour of the digital twin, since the generated traffic reflects real railway operations rather than artificial assumptions.

4.1.4 Simulation

After the generation of the macro-level railway network and the reconstruction of train routes from operational data, the simulation scenario can be executed within SUMO. The objective of this step is to validate the ability of the generated model to reproduce railway movements at a station-to-station level while keeping the simulation computationally manageable.

The macro-level simulation is not intended to reproduce the full Belgian railway network in a single scenario. Although the generated network can theoretically include a large number of stations, simulating the complete national traffic would introduce a high level of complexity and would make the interpretation of the results more difficult. For this reason, the simulation is restricted to selected parts of the network. These parts correspond to chains of neighbouring stations where the train traffic remains moderate. In practice, this means that the user selects a set of stations to be included in the simulation, and the generation pipeline constructs the corresponding local railway subnetwork.

This selection strategy is particularly important for the macro-level model. Since stations are represented as single nodes and complex station layouts are not explicitly modelled, highly congested stations are avoided. Major stations with dense traffic, multiple platforms, many intersecting services, and complex internal routing would require infrastructure details that are not represented at this level of abstraction. By focusing instead on station chains with lower traffic intensity, the simulation remains coherent with the assumptions of the macro-level model. The simplified representation is therefore used in a context where local infrastructure conflicts are less dominant and where station-to-station train movements can still be meaningfully simulated.

A second advantage of this restriction is that real operational trajectories can be directly integrated into the simulation. Since the selected stations form a coherent and connected railway chain, the reconstructed train movements extracted from the punctuality data can be mapped onto the generated SUMO network. Each train route is defined by a sequence of stations and by departure times converted into simulation time. The resulting route file therefore reflects real observed railway operations rather than artificial movements.

The duration of the simulation is configurable. The user can specify the number of operating days to be included in the scenario. This parameter determines the time window used when filtering the operational data and therefore controls the number of train movements inserted into the simulation. A short simulation period can be used for debugging and validation, while a longer period can be selected to analyse traffic behaviour over multiple days.

Once the network, train routes, stop definitions, and simulation configuration files have been generated, the simulation can be launched directly in SUMO. During execution, trains circulate along the station-to-station graph according to the reconstructed timetable. Their movements are mainly governed by the generated routes, departure

times, and travel times associated with the railway connections. The simulation produces standard SUMO output files, including trip information, stop information, and summary statistics. These outputs are then processed in order to obtain structured datasets that can be compared with the original operational data.

Several outputs are produced during simulation execution. The `tripinfo` output contains information regarding completed train journeys, including travel times and delays. The `stopinfo` output records stopping events occurring at stations. Finally, `summary` files provide aggregated information concerning the evolution of the simulation.

The post-processing stage reconstructs both scheduled and simulated arrival and departure times for each train and station. This conversion makes it possible to directly compare simulation results with real railway observations and therefore evaluate the validity of the generated digital twin.

In this way, the macro-level simulation acts as a first validation layer for the proposed railway digital twin. It does not aim to capture all local operational constraints of the railway infrastructure, but it demonstrates that a realistic simulation workflow can be constructed from real station connectivity and punctuality data. It also provides a scalable basis for analysing train movements on selected parts of the railway network before moving toward the more detailed micro-level model.

The complete workflow implemented in the macro-level model can therefore be summarized as follows:

1. Infrastructure and operational data are first preprocessed and standardized.
2. A station-to-station railway network is then generated and converted into a SUMO network.
3. Historical punctuality records are subsequently transformed into train routes.
4. These routes are simulated within SUMO.
5. The resulting outputs are finally converted into datasets suitable for analysis and validation.

4.1.5 Limitations

Although the macro-level model provides a functional and computationally efficient representation of the railway network, it remains limited by its high level of abstraction. Stations are represented as single nodes and railway connections as direct edges, which means that detailed infrastructure elements such as platforms, switches, signals, parallel tracks, and station-level routing constraints are not explicitly modelled. As a result, the model cannot accurately reproduce local operational phenomena such as platform occupation, route conflicts inside stations, or interactions between trains at complex junctions.

Another limitation concerns the scope of the simulation. The macro-level approach is mainly suitable for selected chains of neighbouring stations with moderate traffic intensity. It is therefore less appropriate for dense areas or major stations where the internal infrastructure plays a central role in traffic behaviour. Nevertheless, this simplified model remains useful as a first validation step for the overall digital twin workflow, especially for testing the integration of real operational data into a SUMO-based railway simulation.

4.2 Micro-Level Modeling

While the macro-level model provides a simplified representation of the railway network suitable for large-scale traffic simulations, it does not accurately represent the physical infrastructure required for detailed operational analyses. In particular, elements such as switches, platforms, parallel tracks, and complex station layouts are either simplified or omitted. To overcome these limitations, a micro-level railway network model was developed.

The objective of the micro-level modeling process is to reconstruct the railway infrastructure with a much higher level of spatial and topological accuracy by exploiting detailed infrastructure data together with operational information. Unlike the macro-level model, where stations are represented as single nodes connected through aggregated railway links, the micro-level model explicitly represents every individual track, switch, platform, and operational segment. This detailed representation enables realistic train routing, precise conflict detection, infrastructure capacity analysis, and the simulation of operational scenarios at station level.

The construction of the micro-level network follows a multi-stage processing pipeline. First, raw infrastructure data are collected from the available railway datasets and transformed into a consistent geographic representation. The resulting railway graph is then enriched through several dedicated algorithms responsible for identifying key operational elements of the infrastructure. These include automatic switch detection, platform identification, railway track segmentation, and the construction of a dual graph representation suitable for routing algorithms.

Once the infrastructure has been reconstructed, train trips are generated using either synthetic traffic scenarios or real operational data. These trips are subsequently integrated into the SUMO railway simulation environment, allowing trains to circulate on the generated infrastructure while respecting the physical constraints imposed by the detailed network topology.

Compared to the macro-level model, this approach significantly increases the realism of the generated digital twin. The explicit representation of railway assets enables more accurate train movements, better infrastructure utilization, and the possibility of evaluating local operational phenomena such as conflicts at junctions, platform occupancy, and the propagation of delays within complex stations.

The following sections describe each stage of the micro-level modeling process in detail, starting with the data processing pipeline before presenting the infrastructure extraction algorithms, trip generation methodology, simulation setup, and the main limitations of the proposed approach.

4.2.1 Data Processing Pipeline

The data processing pipeline follows a multi-stage Extract–Transform–Load (ETL) process. Infrastructure data is first collected from the available railway datasets and transformed into a unified format suitable for network reconstruction. The raw geographical information contains track geometries represented as sequences of coordinates describing the railway centerlines.

After data cleaning and normalization, several infrastructure extraction algorithms are applied. The first stage consists of identifying railway switches from the track topology. These switches constitute the connection points between different railway tracks and form

the basis of the network graph.

The second stage focuses on platform detection. Stations are associated with nearby track segments using a proximity-based matching procedure that assigns platforms to the closest available tracks.

Once platforms have been matched, the network is analysed to identify situations where multiple stations are assigned to the same track segment. Such conflicts are resolved through a dedicated track segmentation process that increases the granularity of the infrastructure representation while preserving the original geometry.

The resulting infrastructure model is then converted into a graph representation that can be used for route generation and simulation. Finally, train trips are generated using either synthetic traffic scenarios or operational railway data before being integrated into the SUMO simulation environment.

4.2.2 Infrastructure Modeling

Switch Detection

The railway infrastructure is first represented as a graph in which track extremities correspond to nodes and track sections correspond to edges. For each node, the number of neighbouring track segments is computed.

Nodes connected to exactly two neighbouring segments are considered part of a continuous railway line and are therefore not classified as switches. In contrast, nodes connected to three or more neighbouring segments represent branching or merging points and are identified as switches. These locations correspond to operational infrastructure elements that allow trains to change routes within the network.

The resulting switch and edge datasets forms the foundation of the micro-level network model and is subsequently used for graph construction, route generation, and simulation.

Platform Detection

The infrastructure dataset provides the geographical position of railway stations as well as the number of platforms available at each station and their heights. However, this information is not directly linked to the track segments extracted from the railway network. Consequently, a matching procedure was developed to associate each station platform with the most relevant track segments.

The proposed approach relies exclusively on spatial proximity. For each station, the Euclidean distance between the station location and nearby track segments is computed. The candidate segments are then ranked according to their distance to the station centroid.

Let (N) denote the number of platforms associated with a given station. The algorithm selects the (N) closest track segments and assigns them to the station.

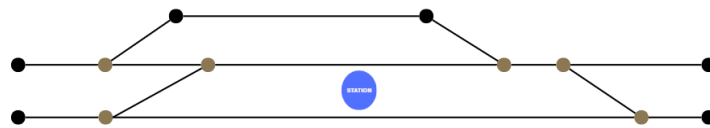


Figure 4.1: Example of platform detection initial setup

For each station, a research zone is defined around its coordinates. This research zone is a circular area with an adaptative radius (e.g., 100 meters) that encompasses the area where they are likely located. The radius of the research zone can be adjusted based on the density of platforms in the area, ensuring that it is large enough to capture all relevant track segments while minimizing the inclusion of irrelevant segments.

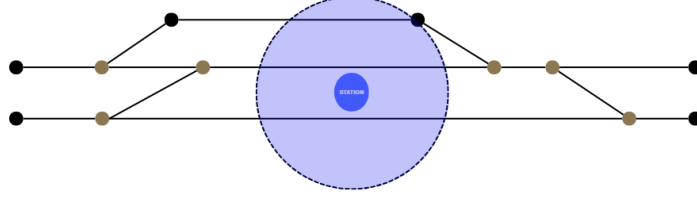


Figure 4.2: Example of platform detection searching zone

This assumption is based on the observation that station platforms are generally located alongside the tracks that serve passenger operations. Therefore, the nearest segments constitute a reasonable approximation of the actual platform locations while requiring only publicly available infrastructure data.

Within the research zone, the script identifies all track segments from the track dataset that intersect with the zone. Then, the script ranks the track segments based on their proximity to the station coordinates. The remaining track segments are classified into three categories (given a station that has n platforms):

1. Best candidates: the track segments that are closest to the station coordinates and are likely to correspond to a platform ($rank \leq n$).
2. Mediocre candidates: the track segments that are within the research zone (or close to it) but are not the best candidate. These segments may correspond to additional platforms or other track segments near the station ($n \leq rank \leq 1.5n$).
3. Irrelevant segments: the track segments that are outside the research zone or are too far from the station coordinates. These segments are unlikely to correspond to platforms and can be disregarded in the platform detection process ($1.5n \leq rank \leq 2n$).

This ranking approach allows the script to prioritize track segments that are most likely to correspond to platforms while minimizing the inclusion of irrelevant segments. By categorizing the track segments based on their proximity to the station coordinates, the script can effectively identify potential platform locations and improve the accuracy of the platform detection process.

The resulting associations are evaluated using distance-based quality indicators. Stations for which the selected segments are located within a short distance are considered to have a high-quality match, whereas larger distances indicate potentially less reliable assignments. This quality assessment is later used to analyse the overall accuracy of the generated infrastructure model.

Track Segmentation

The platform matching process may produce situations in which several stations are associated with the same track segment. Such conflicts occur when a single segment represents

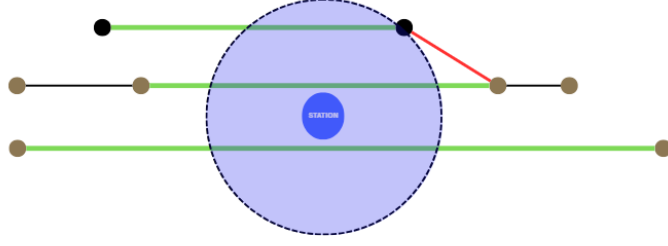


Figure 4.3: Example of platform detection ranking when the number of platform is 3

a long railway section passing through multiple stations, making it impossible to uniquely associate each platform with a distinct operational element.

To resolve this issue, an additional track segmentation procedure is applied after the platform matching stage. The objective is to increase its operational granularity to ensure that each station platform can be uniquely associated with a distinct track segment.

First, all segments assigned to more than one station are identified. Only these conflicting segments are processed, while the remaining network remains unchanged. Each selected segment is then divided into smaller sub-segments according to a maximum segment length threshold of 800 meters.

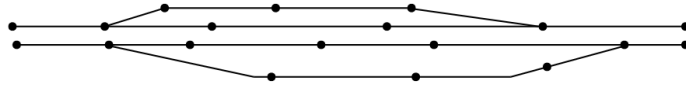


Figure 4.4: Track segmentation Initial setup

The segmentation process follows the existing geometry of the track. New sub-segments are created exclusively using the coordinates already present in the original polyline representation. No interpolation or artificial points are introduced. Distances are computed after projecting the geographical coordinates into the Belgian Lambert 72 coordinate system (EPSG:31370), allowing accurate metric measurements.

For each generated sub-segment, new extremity nodes are created when necessary and the corresponding track attributes are recomputed. The resulting infrastructure therefore preserves the original network geometry while providing a finer representation of the railway infrastructure.

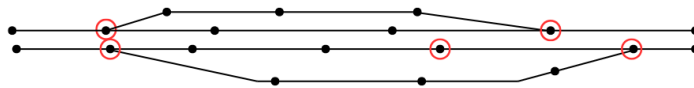


Figure 4.5: Track segmentation applied to a conflicting segment

After segmentation, the platform matching procedure can be repeated on the newly generated segments. The process continues until each station platform is associated with a distinct segment. In practice, a single segmentation iteration was sufficient to eliminate all observed conflicts in the Belgian railway network dataset.

This approach allows the model to maintain a realistic infrastructure representation while ensuring that stations and platforms can be uniquely mapped to simulation entities.

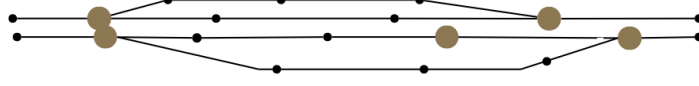


Figure 4.6: Track segmentation results

Such a requirement is particularly important for subsequent routing, timetable generation, and railway traffic simulation tasks.

Dual Graph Construction

Once the railway infrastructure has been segmented into operational track sections, a graph representation is constructed to efficiently compute train routes through the network. The proposed approach adopts a dual graph representation, where each railway track segment becomes a graph node, while the possible transitions between consecutive track segments are represented as directed edges.

This representation considerably simplifies routing operations. Since trains always travel along track sections and move from one section to another through switches or junctions, representing track segments as graph nodes naturally models train movements. Furthermore, routing algorithms operate directly on operational entities instead of geometric infrastructure elements.

The construction of the dual graph is performed in two stages.

First, each track segment is associated with its start and end junctions. A temporary lookup structure is generated in order to quickly retrieve all tracks connected to a given infrastructure node. This preprocessing step accelerates the graph construction by avoiding repeated searches through the complete infrastructure dataset.

Next, the directed graph is created using the **NetworkX** library. For every pair of track segments, a directed edge is added whenever the destination node of the first track corresponds to the origin node of the second track. Formally, given two track segments (t_i) and (t_j) , a directed edge is created if

$$\text{end}(t_i) = \text{start}(t_j). \quad (4.1)$$

The resulting edge represents a valid train movement from one track segment to the next. Each edge is assigned a weight corresponding to the physical length of the originating track segment. Consequently, shortest-path computations minimize the travelled distance across the railway infrastructure.

The resulting graph can be formally defined as $G = (V, E)$, where each vertex $(v \in V)$ represents a railway track segment and each directed edge $((u, v) \in E)$ denotes a feasible transition between two consecutive track segments.

Once the dual graph has been generated, shortest paths between stations can be efficiently computed. Since a station may contain multiple platforms, routing cannot simply be performed between station identifiers. Instead, every platform of the origin station is paired with every platform of the destination station. For each platform pair, the corresponding track segments are identified and Dijkstra's shortest-path algorithm, provided by NetworkX, is executed on the dual graph using the track lengths as edge weights.

The algorithm returns the ordered sequence of track segments connecting the two platforms. This sequence is then converted into the corresponding SUMO edge identifiers

by identifying the physical connections between consecutive track segments. The resulting edge sequence can be directly inserted into SUMO route definitions without requiring any additional processing.

To avoid repeatedly solving identical routing problems, all computed routes are stored in a cache indexed by the origin station, destination station, and platform pair. Subsequent requests for the same station pair immediately retrieve the previously computed route, significantly reducing the computational cost during trip generation. Finally, whenever the operational dataset contains trips between station pairs that have not yet been processed, the corresponding routes are computed on demand and added to the cache. This strategy guarantees that every station-to-station movement required by the timetable is associated with a valid infrastructure path before the simulation is generated.

4.2.3 Trip Generation

The next step consists of generating train trips that will be executed during the simulation. A trip defines the movement of a train through the railway network by specifying its origin, destination, departure time, and the route computed between these locations. The quality and realism of the generated trips directly influence the representativeness of the simulation and, consequently, the validity of the obtained results.

Different strategies can be adopted for trip generation. Synthetic trips are useful for validating the generated infrastructure, evaluating routing capabilities, and creating controlled traffic scenarios with varying levels of congestion. Conversely, trips reconstructed from real operational data allow the simulation to reproduce actual railway operations and provide a more realistic representation of traffic conditions.

To address both requirements, this work implements two complementary trip generation approaches. The first approach automatically generates random trips over the reconstructed railway network, while the second reconstructs train movements from real operational datasets. The following sections present these two methodologies in detail.

Random Trips

The first trip generation strategy aims at producing synthetic railway traffic over the reconstructed network without relying on historical operational data. This approach is primarily intended to validate the generated infrastructure, verify the connectivity of the network, and evaluate the routing capabilities of the simulation under controlled conditions.

To achieve this, the reconstructed train paths obtained during the infrastructure generation phase are first analyzed in order to identify every station that appears either as the starting point or the terminal point of an existing railway service. These stations constitute the candidate locations from which synthetic trips can be generated.

Once the candidate stations have been identified, weighting files are automatically generated for SUMO's `randomTrips.py` utility. Each platform associated with a candidate station receives an identical weight, while all other tracks are excluded from the selection process. Consequently, the trip generator only selects railway platforms corresponding to valid departure and arrival stations, preventing the creation of unrealistic trips originating from intermediate tracks, switches, or maintenance sections.

The `randomTrips.py` tool is then executed on the generated railway network. The generation process is constrained to railway edges only, ensuring that all produced trips are compatible with the rail infrastructure. Several parameters are used to increase the

realism of the generated traffic. In particular, a minimum travel distance is imposed between the origin and destination to avoid trivial journeys, while a variable insertion rate is specified throughout the simulation period to emulate fluctuations in railway traffic intensity during the day. The resulting output is a set of synthetic trips, each containing a departure edge, an arrival edge, and an associated departure time.

Although these trips are syntactically valid, they do not necessarily correspond to railway services that can actually be operated within the Belgian railway network. To ensure operational consistency, an additional validation step is performed. For each generated trip, the departure and arrival platforms are first mapped back to their corresponding stations. The algorithm then verifies whether these two stations belong to at least one reconstructed train path extracted from the operational dataset. Only trips whose origin and destination are connected by an existing railway service are retained. Trips failing this verification are discarded.

Finally, the validated trips are converted into a SUMO route file using the `duarouter` routing engine. This tool computes the complete railway path through the generated infrastructure while respecting the network topology and railway connectivity constraints. The resulting routes are subsequently used as input for the simulation.

Real Data Trips

In addition to synthetic traffic generation, a second approach was developed to reproduce actual railway operations using real punctuality data. Unlike the previous method, which generates artificial train movements, this approach aims at recreating real train services by preserving their scheduled sequence of stops and planned timetable. The objective is to obtain a simulation that closely reflects real railway traffic and therefore provides a more realistic digital twin of the Belgian railway network.

In principle, this approach would allow all train journeys present in the selected operational dataset to be inserted into the simulation. For each train and each operating day, the timetable is reconstructed as an ordered list of station stops. Each station is then associated with one of its detected platforms, and the route between consecutive stations is computed using the dual graph constructed during the infrastructure modeling phase. The final route of a train is obtained by concatenating all the paths computed between successive stops.

The first step consists in processing the punctuality dataset in order to reconstruct each train service individually. The operational records are first cleaned by removing incomplete entries and converting the planned arrival and departure timestamps into simulation time expressed in seconds from the beginning of the simulation. The records are then grouped according to the train identifier and operating date, allowing each train journey to be reconstructed as an ordered sequence of stations associated with their planned arrival and departure times.

Once the timetable has been reconstructed, each station visited by a train is associated with the set of platforms detected during the infrastructure modeling phase. Platform assignment is then performed while respecting the timetable constraints and ensuring that no two trains occupy the same platform simultaneously. A configurable safety buffer is introduced between successive platform occupations in order to prevent unrealistic conflicts and better represent operational constraints encountered in real railway systems.

The main difference with the random trip generation strategy lies in the route computation. Instead of relying on SUMO to determine a shortest path between two stations, this methodology exploits the dual graph introduced in Section 4.2.2. For every pair of

consecutive stations in a train service, the corresponding path is searched within the dual graph, taking into account the selected departure and arrival platforms. The complete train route is then obtained by concatenating the paths computed between every pair of successive stations. This approach guarantees that trains follow the detailed railway topology reconstructed during the infrastructure modeling stage and remain consistent with the operational characteristics of the network.

However, although the methodology is conceptually sound, the current implementation did not produce exploitable simulation results. During the platform assignment process, many consecutive station pairs could not be connected because no valid path was found in the generated dual graph. As a consequence, a large proportion of train services were rejected before route generation, preventing the construction of a complete operational timetable.

The absence of feasible paths does not originate from the timetable itself but rather from the current state of the generated dual graph. Some infrastructure elements or connections required to guarantee complete network connectivity are still missing or incorrectly represented, making certain station-to-station journeys impossible despite their existence in the real railway network. Improving the connectivity and completeness of the dual graph therefore constitutes a necessary step before realistic timetable-based simulations can be successfully performed.

Despite these limitations, this approach establishes the foundations for integrating real operational data into the proposed railway digital twin. Once the dual graph reconstruction is sufficiently complete, the methodology will enable the simulation of real train movements while preserving both the actual timetable and the detailed infrastructure topology.

4.2.4 Simulation

Once the micro-level infrastructure has been generated and train trips have been created, the resulting scenario can be simulated in SUMO. In contrast with the macro-level model, the micro-level simulation relies on a detailed representation of the railway infrastructure. Tracks, switches, platforms, and operational segments are explicitly represented, which allows trains to circulate through a much more realistic network topology.

The simulation procedure is designed to support two different types of traffic scenarios. The first scenario uses randomly generated trips, while the second attempts to reproduce real train movements extracted from operational data. These two approaches serve different purposes. Random trips are mainly used to validate the generated infrastructure and to test whether trains can move through the network under controlled conditions. Real-data trips, on the other hand, aim to reproduce actual railway operations and therefore provide a more realistic simulation of the Belgian railway network.

Random-trip simulation

The random-trip simulation is based on synthetic train movements generated from the reconstructed railway network. Although the generated trips do not correspond to real train services, the generation process is constrained in order to make them as realistic as possible. The objective is not to reproduce a specific timetable, but to create plausible railway movements that can be executed on the micro-level infrastructure.

Additional parameters provided by SUMO's trip generation and routing tools are then adjusted to improve the realism of the scenario. The insertion rate controls the frequency

at which trains are introduced into the simulation. By modifying this parameter, it is possible to generate different traffic densities and to emulate periods with more or fewer train movements. Other parameters, such as the minimum trip distance, the departure time distribution, and the selection weights assigned to candidate edges, are also used to avoid unrealistic routes and to favour meaningful station-to-station movements.

After the random trips have been generated, they are routed through the micro-level network. The routing step computes complete paths between the selected departure and arrival platforms while respecting the topology of the SUMO network. The resulting routes are then inserted into the simulation configuration. This process makes it possible to test whether the reconstructed infrastructure is sufficiently connected and whether the generated railway assets can support train circulation.

The number of simulated days is configurable by the user, as in the macro-level model. This parameter controls the simulation horizon and influences the number of generated trips. A short duration is useful for validating the network and detecting routing errors, while a longer duration can be used to evaluate the behaviour of the infrastructure under more sustained traffic conditions.

The random-trip simulation therefore plays an important validation role. Since the generated trips are not constrained by a real timetable, they provide a flexible way to test the robustness of the network, the correctness of the platform detection process, the quality of the routing, and the ability of SUMO to execute train movements on the reconstructed infrastructure.

Real-data simulation

The second simulation scenario aims to use real operational data instead of synthetic trips. In this case, the objective is to reproduce the train services contained in the punctuality dataset. Each train movement should be reconstructed from the real sequence of stations visited by the train, associated with the corresponding planned arrival and departure times, and then mapped onto the detailed micro-level infrastructure.

However, the current implementation of this scenario did not produce exploitable simulation results. The main difficulty appears during the mapping of real train movements onto the reconstructed micro-level infrastructure. For majority of station pairs, no valid path could be found in the dual graph. As a consequence, a significant number of train services were rejected before being converted into SUMO routes, preventing the generation of a complete real-data simulation scenario.

This issue does not necessarily indicate that the operational data is incorrect. Rather, it highlights limitations in the current state of the reconstructed micro-level graph. Some connections required to reproduce real railway movements are missing, incorrectly oriented, or not represented with sufficient topological consistency. Since the real-data simulation requires complete paths between all consecutive stations of each train journey, even small connectivity gaps can prevent an entire service from being simulated.

Despite this limitation, the real-data simulation workflow remains an important contribution of the implementation. It defines the structure required to integrate operational railway data into the digital twin and identifies the main technical obstacle that must be solved in future work. Improving the completeness of the dual graph, refining the representation of switches and track directions, and strengthening the platform assignment strategy should make it possible to execute real timetable-based simulations in a future version of the system.

4.2.5 Limitations

The micro-level model offers a more detailed representation of the railway infrastructure, but this additional realism also introduces several limitations. The quality of the simulation strongly depends on the completeness and correctness of the reconstructed infrastructure graph. Missing connections, incorrect track orientations, imperfect switch detection, or inaccurate platform assignments can prevent the computation of valid routes and may lead to unrealistic or incomplete simulations.

A second limitation concerns the integration of real operational data. While random trips can be generated and simulated more easily, the use of real train trajectories requires a continuous and valid path between every pair of consecutive stations in the timetable. In the current implementation, this requirement could not always be satisfied, which prevented the generation of a fully exploitable real-data simulation. These limitations are only introduced here briefly, as they are discussed in more detail in the dedicated limitations chapter.

Chapter 5

Results

This chapter presents the results obtained from the modeling and simulation pipeline developed in this thesis. The evaluation focuses on two main aspects. First, the generated datasets are compared with the available reference data in order to assess their completeness. Second, the infrastructure elements extracted during the modeling process are analysed to evaluate the quality of the generated railway representation.

The results are presented separately for the macro-level and micro-level approaches when necessary. This distinction is important because the two models do not represent the railway network at the same level of detail.

5.1 Dataset Validation

5.1.1 Station Coverage

Station coverage evaluates the number of stations detected and integrated into the generated datasets. This metric is important because stations are central elements in both modeling approaches. In the macro-level model, stations are the main nodes of the graph. In the micro-level model, stations are used to associate platforms and tracks with operational stopping points.

Simulation Level	Generated Stations	Reference Stations	Coverage (%)
Macro-Level	559	559	100
Micro-Level	553	554	99.82

Table 5.1: Station coverage for macro-level and micro-level simulations

The results show that station coverage is very high for both modeling approaches. For the macro-level simulation, all 559 stations present in the reference dataset are generated, corresponding to a coverage of 100.00%. This indicates that, at the station-to-station level, the macro-level generation process is able to fully preserve the set of stations required for the generated network.

For the micro-level simulation, 553 stations are generated from a reference dataset containing 554 stations, corresponding to a coverage of 99.82%. This result is also very strong, especially considering that the micro-level model requires a more detailed association between stations, platforms, and physical track infrastructure. The small difference

between the reference and generated datasets may be explained by data inconsistencies, preprocessing filters, or the absence of a valid platform-track association for one station.

Overall, the station coverage results confirm that the generated datasets provide a reliable station basis for the rest of the modeling pipeline. The macro-level approach achieves complete station coverage, while the micro-level approach remains almost complete despite the additional infrastructure constraints introduced by the detailed model.

5.1.2 Platform and Track Coverage

The second part of the dataset validation concerns platform and track coverage. These elements are especially important for the micro-level model, where the objective is to represent the railway infrastructure in more detail. Platforms define where trains can stop inside stations, while tracks define the physical railway segments used for circulation.

Category	Reference Data	Simulation Data	Coverage (%)
Platforms	1551 platforms	1560 platforms	99.42
Tracks (Macro-Level)	1385 tracks	1560 tracks	100
Tracks (Micro-Level)	25028 tracks	26721 tracks	106.59

Table 5.2: Platform and track coverage for micro-level simulation

The platform coverage obtained for the micro-level model is also very high. The reference dataset contains 1560 platforms, while the generated dataset contains 1551 platforms, resulting in a coverage of 99.42%. This shows that the platform detection and matching process is able to identify almost all platforms available in the reference data. However, it does not guarantee that the generated platforms are correctly positioned on the tracks, which is a separate aspect of the infrastructure modeling process.

Track coverage differs between the macro-level and micro-level models. In the macro-level simulation, the reference data contains 1385 station-to-station tracks and the generated dataset also contains 1385 tracks, giving a coverage of 100%. This result is expected because the macro-level model directly represents connections between stations and does not attempt to reconstruct the detailed internal railway topology.

For the micro-level simulation, the generated dataset contains 26721 tracks, while the reference dataset contains 25068 tracks. This corresponds to a coverage of 106.59%. A value higher than 100% does not necessarily indicate an error. It reflects the fact that the micro-level generation process may create more detailed or segmented track representations than those present in the reference dataset. In particular, a single reference track can be represented by multiple generated segments after preprocessing, switch detection, and track segmentation.

Overall, the platform and track coverage results show that the generated infrastructure is highly complete. However, the interpretation of track coverage must take into account the difference between reference tracks and simulation-ready SUMO edges. The generated micro-level network is not only a copy of the reference data, but a transformed representation adapted to routing and simulation constraints.

5.2 Infrastructure Modeling Results

After validating the general completeness of the datasets, the next step is to evaluate the results of the infrastructure modeling process. This part focuses on the railway elements that were extracted, transformed, or reconstructed during the implementation. The objective is to assess whether the generated infrastructure provides a coherent basis for railway simulation.

The infrastructure modeling results confirm the high level of coverage already observed in the dataset validation step. Stations are almost completely detected in both modeling approaches, and platforms are also recovered with a high success rate in the micro-level model. Switches require a more nuanced interpretation, because they are inferred from the generated railway topology rather than directly copied from the reference dataset.

Category	Reference Data	Simulation Data	Detection (%)
Stations (Macro-Level)	559	559	100
Stations (Micro-Level)	554	553	99.82
Platforms (Micro-Level)	1560	1551	99.42
Switches (Micro-Level)	10379	12343	118.92

Table 5.3: Detected stations, platforms and switches for macro-level and micro-level simulations

The detection percentage reaches 100.00% for macro-level stations and 99.82% for micro-level stations. Platform detection reaches 99.42%, which confirms that most platforms could be matched to the generated infrastructure. These results are important because platforms are used as stopping locations for trains and as origin or destination points during trip generation.

Switch detection provides a different type of result. The reference data contains 10379 switches, while the generated micro-level infrastructure contains 12343 detected switches, corresponding to a coverage of 118.92%. As with micro-level track coverage, this value above 100% must be interpreted carefully. It does not simply mean that the model detects more correct switches than expected. Instead, it indicates that the algorithm identifies more candidate switch points than the number of switches available in the reference dataset.

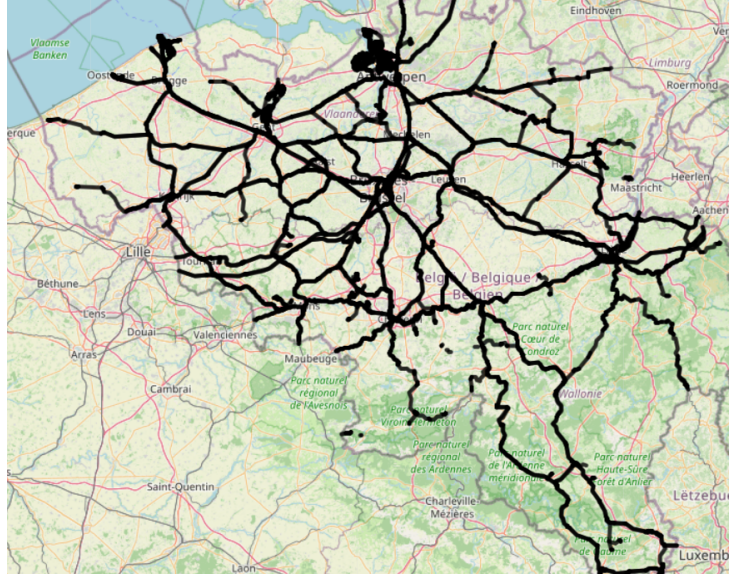


Figure 5.1: Network before and after switch detection.

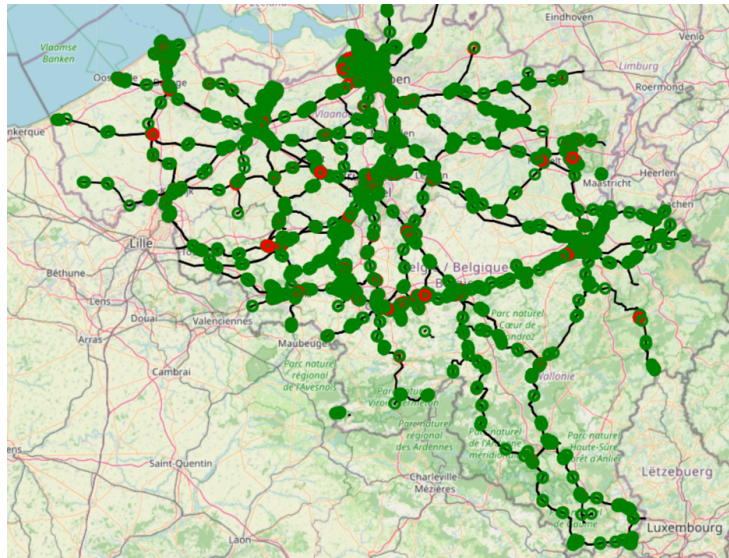


Figure 5.2: Network before and after switch detection (Green dots represent detected switches with 3 connections and Red dots represent detected switches with 4 or more connections)

This visual comparison makes it possible to understand how the algorithm enriches the infrastructure by identifying branching and merging points across the network.

This difference can be explained by the way switches are inferred from the railway topology. In the generated network, a switch is detected when a node connects more than two track segments and therefore behaves as a branching or merging point. However, this topological definition may identify some elements that are not classified as switches in the reference data. It may also split complex junctions into several smaller switch-like elements.

The visual comparison of the network before and after switch detection helps illustrate this effect. The switch detection process enriches the generated infrastructure by identifying relevant branching points and making the network more suitable for routing. In

dense areas such as Bruxelles-Midi, the detected switches provide a much more detailed representation of the railway topology, but they also highlight the difficulty of matching generated topological elements exactly with reference infrastructure objects.

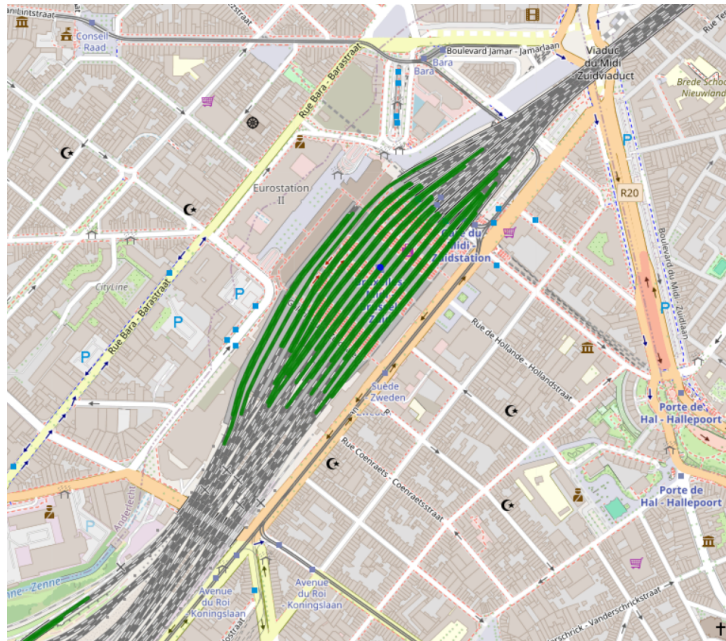


Figure 5.3: Platform matching result on Bruxelles-Midi

This station is particularly relevant because it contains a dense set of tracks and platforms, making it a useful example for evaluating whether the platform matching algorithm can handle complex station environments.

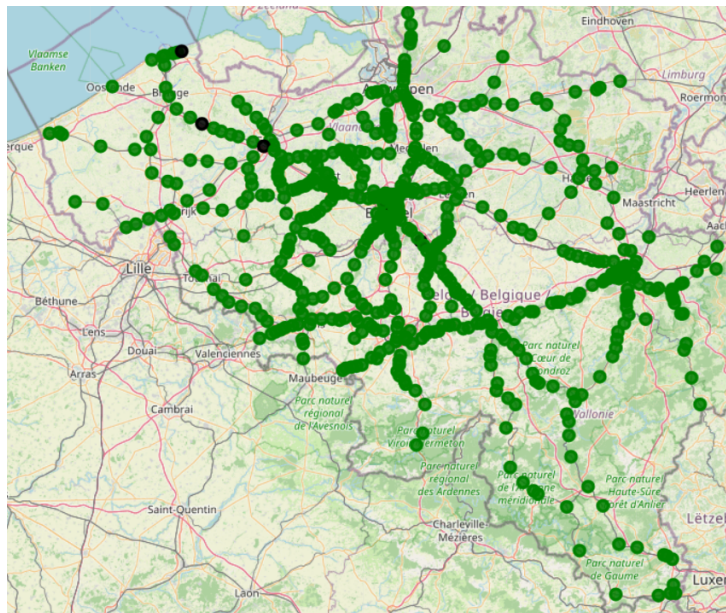


Figure 5.4: Platform matching quality map

The platform matching figures further illustrate the quality of the generated infrastructure. They show how platforms are associated with the corresponding railway tracks

and make it possible to visually inspect the quality of the matching process. This qualitative validation complements the numerical coverage metrics and helps identify local cases where the matching may be less accurate.

TUBIZE Station_ID: 1160.0 n quais: 3 n candidats: 8 rayon utilisé: 100 m min distance: 0.03 m mean distance: 45.14 m qualité: good Good quality	WEERDE Station_ID: 1224.0 n quais: 4 n candidats: 10 rayon utilisé: 100 m min distance: 32.24 m mean distance: 57.79 m qualité: warning Ok quality	HANSBEKE Station_ID: 518.0 n quais: 4 n candidats: 2 rayon utilisé: 300 m min distance: 13.43 m mean distance: 15.45 m qualité: insufficient Insufficient quality
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Figure 5.5: Platform matching quality examples

The platform matching quality is further illustrated by a qualitative comparison of several representative stations. As shown in Figure 5.5, the matching process can produce different levels of quality depending on the local infrastructure configuration. For instance, Tubize is classified as a good-quality match, with a very small minimum distance between the detected platform and the candidate track, which suggests a reliable association. Weerde corresponds to an intermediate case, labelled as a warning, where the matching remains acceptable but presents a larger average distance, indicating a less precise association. By contrast, Hansbeke is classified as insufficient quality, showing that the algorithm may encounter difficulties in some local configurations, especially when the search radius is larger or when the number of candidate tracks does not allow a clear correspondence to be established.

5.3 Simulation Results

5.4 Computational Performance

5.5 Discussion

Chapter 6

Limitations

Chapter 7

Conclusion

7.1 Summary of Contributions

7.2 Key Findings

7.3 Practical Implications

7.4 Future directions

The work presented in this thesis provides a first implementation of a railway digital twin based on real infrastructure and operational data. However, several extensions could be considered in order to transform the current simulation framework into a more complete decision-support tool for railway operators and researchers. This section presents four future research directions that naturally follow from the proposed macro-level and micro-level modeling approaches.

7.4.1 Counterfactual Simulation and What-If Scenarios

A first direction for future work concerns the development of counterfactual simulations and what-if scenarios. In the current implementation, the simulation primarily aims to reproduce railway operations based on existing infrastructure and, when possible, real train trajectories. A counterfactual extension would go one step further by allowing the user to modify specific elements of the system and observe how these modifications affect the simulated railway traffic.

In practice, this could consist of testing alternative operational or infrastructure configurations. For example, a user could simulate the closure of a station, the removal of a track section, a change in train frequency, the introduction of additional services, or the modification of departure times. The objective would not only be to reproduce what happened in the real network, but also to evaluate what could have happened under different assumptions.

This direction could be implemented by extending the current scenario generation pipeline. Since the macro-level model already represents stations and connections as a graph, it could be used to rapidly test high-level disruptions or modifications in station-to-station connectivity. The micro-level model, on the other hand, could support more detailed counterfactual experiments, such as the unavailability of a platform, a switch,

or a track segment. In both cases, the simulation configuration could include a scenario layer defining which infrastructure elements are modified, removed, or constrained before the simulation is executed.

The main utility of this extension would be to transform the digital twin into an exploratory tool. Railway operators could use it to evaluate the consequences of operational decisions before applying them in the real network. Researchers could also use it to compare different traffic management strategies under controlled and reproducible conditions. This would make the framework useful not only for simulation, but also for planning, resilience analysis, and decision support.

7.4.2 Tracks Works Planning

A second future direction concerns the use of the digital twin for track works planning. Railway maintenance and infrastructure works often require temporary closures, reduced capacity, speed restrictions, or rerouting strategies. These interventions are necessary to preserve infrastructure quality, but they can also generate significant operational constraints. A simulation-based tool could therefore help evaluate the impact of planned works before they are carried out.

In the context of this thesis, this extension would be particularly relevant because the generated network already contains infrastructure elements that can be selectively modified. At the macro level, track works could initially be represented by removing or penalizing railway connections between stations. This would make it possible to evaluate whether trains can still be routed through alternative paths and to estimate the effect of the works on the global connectivity of the selected network area.

At the micro level, track works could be modeled with a higher level of detail. Specific track segments, switches, or platforms could be marked as unavailable during a given time window. The route generation and simulation process would then need to take these constraints into account. This could be implemented by adding temporal availability attributes to the infrastructure graph and by adapting the routing algorithm so that it avoids unavailable elements when computing train paths.

The main benefit of this direction is operational. A digital twin capable of simulating planned works could help identify bottlenecks, anticipate rerouting needs, and compare several maintenance schedules. It could also support the evaluation of different intervention strategies, for example by comparing a short but highly disruptive closure with a longer intervention affecting fewer services. In this way, the framework could contribute to a better balance between infrastructure maintenance and service continuity.

7.4.3 Real-Time Forecasting and Incident Propagation

A third future direction is the integration of real-time forecasting and incident propagation mechanisms. The current implementation mainly focuses on reconstructing and simulating railway operations from static or historical datasets. However, a railway digital twin becomes more powerful when it can continuously integrate new operational information and use it to predict the evolution of the network state.

This extension would consist of using live or near-real-time data to update the simulation state. For example, if a train is delayed at a given station, the digital twin could estimate how this delay may propagate to subsequent stations, to other trains, or to shared infrastructure sections. The objective would be to move from retrospective simu-

lation toward predictive simulation, where the system helps anticipate future disruptions before they fully materialize.

This direction could be implemented progressively. A first step would be to use the existing punctuality data as historical input to train statistical or machine learning models capable of estimating delay propagation patterns. These models could then be connected to the simulation pipeline. The macro-level model could be used to study propagation at the station-to-station level, while the micro-level model could be used to analyse how local infrastructure conflicts contribute to the spread of delays. In a more advanced version, the simulation could be updated dynamically when new train positions, delays, or incidents are received.

The utility of this extension is particularly important for operational monitoring. A digital twin with forecasting capabilities could help identify the stations or track sections most likely to be affected by an ongoing incident. It could also support proactive traffic management by testing corrective actions, such as rerouting, rescheduling, or temporary prioritization of certain trains. This would make the digital twin not only a representation of the railway system, but also a tool for anticipating and mitigating disruptions.

7.4.4 Framework Generalization

A final future direction concerns the generalization of the proposed framework. The implementation developed in this thesis is based on Belgian railway data and on the specific structure of the available infrastructure and operational datasets. While this makes the work relevant for the considered case study, a more general framework would allow the same methodology to be applied to other railway networks, datasets, or simulation contexts.

Generalization would require separating the core modeling logic from the data-specific processing steps. In the current implementation, several transformations are adapted to the format and assumptions of the available datasets. A future version of the framework could introduce a more modular architecture, where each data source is handled by a dedicated adapter. These adapters would transform external datasets into a common internal representation of stations, tracks, switches, platforms, and train movements.

Such a modular design would also make it easier to support different levels of detail. Depending on the available data, the framework could generate either a macro-level model, a micro-level model, or a hybrid representation combining both. This would be useful because railway data availability varies significantly between countries, operators, and infrastructure managers. Some networks may provide detailed track-level data, while others may only provide station-level connectivity and timetable information.

The main utility of framework generalization would be to increase the scientific and practical value of the work. From a research perspective, it would make the approach easier to reproduce and compare across different railway networks. From an operational perspective, it would allow the framework to be adapted to new contexts without rewriting the entire pipeline. Ultimately, this would move the project from a case-specific implementation toward a reusable methodology for railway digital twin generation and simulation.

7.5 Final Remarks

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